Yifan Mo

Quantitative Research & Engineering | Program Analysis & AI Agents

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Quantitative Research & Engineering

End-to-end research stack. Built and owned a research-to-execution stack screening approximately **1,000 listed perpetual contracts across Binance, Hyperliquid, and Aster** (Sep. 2026 snapshot).

Research and validation. Cross-sectional factor research and portfolio construction; market-regime splits, parameter perturbation, stability checks, and cross-venue validation.

Engineering and risk. Python; market-data pipelines; backtesting and live trading; exchange integration; order-book-aware execution; position, liquidity, funding-rate and tail-risk controls; live-to-backtest reconciliation.

Research focus: Program analysis, software security, runtime verification, and AI agents for understanding complex software systems.

Education

Sun Yat-sen University Sep. 2022 - Jun. 2027 (expected)

Ph.D. candidate in Software Engineering. Advisor: Prof. Zibin Zheng.

South China University of Technology Graduated in 2022

B.Sc., Information and Computational Science; B.Eng., Computer Science and Technology.

Research Experience

The Hong Kong University of Science and Technology Apr. 2026 - Present

Research Assistant, Cybersecurity Lab. FinTech and computer-security research.

Ant Group, Skyward Lab Jun. 2025 - Apr. 2026

Research Intern

DeFi and blockchain-security research; security-review support for cross-chain infrastructure and stablecoin systems.

State Grid Doctoral Workstation 2024

Visiting Researcher

Smart-contract security research and auditing support for blockchain-based electricity trading.

Selected Publications

Revealing Honeypots in High-Frequency Interactions on Decentralized Applications

First author. IEEE Transactions on Information Forensics and Security, 2026.

Analyzed **238M+ Ethereum transactions**; identified **636 attack events affecting 99 contracts** with **94.6% detection precision**, and recovered **82.4% of failed-transaction semantics**.

DOI: 10.1109/TIFS.2026.3687759

Toward Automated Detecting Unanticipated Price Feed in Smart Contract

First author. ACM International Symposium on Software Testing and Analysis (ISSTA), 2023.

Built VeriOracle runtime verification; evaluated **500K+ transactions across 13 exploited DeFi protocols**, with **3.85-second average verification latency** in the evaluated sample.

DOI: 10.1145/3597926.3598133

Selected Systems, Community and Service

OWL - Agent-Assisted Trading Workflows. Built and deployed on-chain semantic analysis workflows with deterministic position, execution and risk controls; implemented correction, alerts, reporting and human review.

Agora OS. Co-designed an agentic decision and execution architecture; implemented signal processing, portfolio construction, execution, and risk controls.

Community and open source. Founded Blockspace and led approximately **10 developers building 4 tools**; contributed **2 merged PRs** to DeFiHackLabs with Venus donation-attack PoCs (#1027, #1030); published **57 technical posts** on quantitative research and trading systems at Quantclass.

Academic service: BlockSys 2023 PC; reviewer / sub-reviewer: CVPR, ICML, IEEE TSC, IEEE TSMC, ACM TOSEM.